

Madrid demographic residual

Current research results and audit reports

2 October 2026

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Observed research results and validation

Updated 2 October 2026. These are research results from observed municipal data, not an official statistical release. The official release gates remain closed.

City results

Baseline ages 0–89 are followed to ages 10–99. National INE table 27153 annual sex-specific risks are compounded along each ageing cohort. The same sex-specific schedule is applied to ESP and EXT. Ages born during the interval and endpoint ages 100+ are shown separately.

Window	Eligible baseline	Expected survivors	Observed endpoint	Residual	Per 1,000 expected
2015-2025	3,109,701	2,827,032.405	3,275,658	448,625.595	158.691
2014-2024	3,138,522	2,854,624.882	3,207,983	353,358.118	123.784

The 2015–2025 result excludes 262,476 endpoint children aged 0–9 and 2,230 people aged 100+. One baseline record with age 135 is excluded from analytical composition and remains in source accounting. A residual is a stock difference; it cannot separate migration, registration changes, nationality reclassification or model error.

Independent survival check

A separate implementation propagates each population count annually with 40-digit Decimal arithmetic using published survivor ratios $l[x+1,y] / l[x,y]$. It does not call the production qx survival helper. Every eligible age/sex/nationality cell passes a bound derived from the six-decimal precision of the published qx-per-thousand and lx values.

Window	Cells checked	Maximum cell difference	Total expected difference	Total rounding bound
2015-2025	360	0.00005068	0.00001781	0.01591917
2014-2024	360	0.00004784	0.00015961	0.01606745

Mortality and terminal-age sensitivity

Terminal cutoffs change the cohorts being compared. Compare Madrid and national mortality at the same endpoint cutoff of 95. The Madrid variant scales the national single-age hazard pattern so the product of annual survival within each closed age band reproduces the published Madrid band survival. This is an approximation, not an observed single-age Madrid schedule. Open ages are never expanded.

Window	Variant	Endpoint ages	Expected	Observed	Residual
2015-2025	Spain under90	10–89	2,767,820.4	3,215,476	447,655.6
2015-2025	Spain under95	10–94	2,813,585.8	3,262,065	448,479.2
2015-2025	Spain under100	10–99	2,827,032.4	3,275,658	448,625.6
2015-2025	Spain half-age under100	10–99	2,814,985.3	3,275,658	460,672.7
2015-2025	Spain lx under100	10–99	2,827,032.4	3,275,658	448,625.6
2015-2025	Madrid approximation under95	10–94	2,838,073.9	3,262,065	423,991.1

Window	Variant	Endpoint ages	Expected	Observed	Residual
2014–2024	Spain under90	10–89	2,797,219.4	3,150,151	352,931.6
2014–2024	Spain under95	10–94	2,841,968.0	3,195,405	353,437.0
2014–2024	Spain under100	10–99	2,854,624.9	3,207,983	353,358.1
2014–2024	Spain half-age under100	10–99	2,842,542.2	3,207,983	365,440.8
2014–2024	Spain lx under100	10–99	2,854,624.9	3,207,983	353,358.1
2014–2024	Madrid approximation under95	10–94	2,865,779.9	3,195,405	329,625.1

At endpoint ages 10–94, the 2015–2025 national residual is 448,479.2 and the Madrid approximation is 423,991.1 (24,488.1 fewer). Half-age interpolation at ages 10–99 raises the national residual by 12,047.1. These are scenario differences, not confidence intervals. The denominator threshold remains 20 expected survivors; rates below it are suppressed.

Annual reconciliation and intermediate years

Eleven of the twelve January section files exactly match all 400 city age/sex/nationality cells and all four full-population monthly controls. Raw blanks remain unchanged. For matched years only, research aggregation conditionally treats blanks as zero under the assumptions of the same complete universe, nonnegative counts, correct known cells and no compensating errors.

Year	Raw known total	Raw blank cells	Controls match	Invalid-age people
2014	3,176,508	296,007	Yes	0
2015	3,149,663	304,657	Yes	1
2016	3,174,945	302,903	Yes	0
2017	3,191,117	300,262	Yes	0
2018	3,231,062	297,248	Yes	0
2019	3,275,195	289,584	Yes	0
2020	3,345,894	280,036	Yes	0
2021	3,292,949	0	No	1
2022	3,296,033	279,395	Yes	1
2023	3,352,448	0	Yes	0
2024	3,473,648	0	Yes	0
2025	3,540,364	0	Yes	0

2021 fails reconciliation. The section file totals 3,292,949; the monthly bank totals 3,327,119, a gap of 34,170. There are 394 age-control discrepancies. No local zero inference is approved for that year. The intermediate-year city calculation uses the bank’s explicit city age/sex/nationality counts, with an explicit source-change flag. This discrepancy remains an empirical release blocker.

The intermediate comparison follows a fixed baseline population aged 0–89 in 2015. Endpoint ages increase each year; no cohorts are silently added to the baseline.

Endpoint	Eligible observed	Expected	Residual
2016	3,116,241	3,085,989.6	30,251.4
2017	3,109,084	3,062,020.8	47,063.2
2018	3,126,292	3,036,343.6	89,948.4
2019	3,148,020	3,009,532.5	138,487.5
2020	3,194,837	2,982,500.9	212,336.1
2021	3,157,725	2,949,673.0	208,052.0
2022	3,110,181	2,919,166.0	191,015.0
2023	3,144,355	2,887,436.0	256,919.0
2024	3,236,410	2,857,330.4	379,079.6
2025	3,275,658	2,827,032.4	448,625.6

Spatial comparability and small-area results

The research overlay contains 2,343 common section zones: 2,221 unchanged and 122 changed components. Every whole section maps once with weight 1. The maximum outer-union symmetric difference ratio is 0.000000143, below the predeclared 0.001 tolerance. No population is allocated by land area.

All parent codes attached to each common zone are linked across the two vintages. Connected parent groups are pooled in full, yielding 123 common barrio areas and 20 common district areas. This retains ambiguous memberships instead of choosing a majority. These are derived research aggregates, not approved historical administrative boundaries.

The pooled district is Hortaleza + Barajas. Pooled barrio groups are Valdefuentes + Aeropuerto + Casco Histórico de Barajas + Timón + Corralejos; Casco Histórico de Vallecas + Ensanche de Vallecas; and all four Vicalvaro barrios. Stable parent-code groups are still labelled as common research areas because independent dated parent boundaries have not been verified.

Spatial results omit 12 people with unmapped section codes in 2015 and 5 in 2025; one additional baseline person has invalid age 135. The mapped eligible baseline is 3,109,689, expected 2,827,020.973, actual 3,275,653 and residual 448,632.027. The whole-city result includes the exceptional geography and is available separately.

All section, common barrio, common district and mapped-city results pass sex/nationality sums, survivor bounds, stock accounting and additive hierarchy checks. Download the area inventory in [research-area-results.csv](#). Full cohort detail, composition and age-band outputs are packaged in the explorer.

Source attribution and licensing

Population: Ayuntamiento de Madrid, municipal padrón January snapshots 2014–2025 and Banco de Datos Municipal January city controls, retrieved 2 October 2026. Mortality and census-section geometry: Instituto Nacional de Estadística (INE), national table 27153, Madrid closed bands in table 27154, census sections 2015/2025.

The municipal padrón catalogues explicitly declare CC BY 4.0. The archived INE reuse terms declare CC BY 4.0 as the general statistical-information licence absent a specific exception. The source lock records evidence for the inputs used in research; unresolved licence metadata for other audit candidates is not silently filled. Transformations include conditional zero aggregation, cohort survival, common-zone unions and display simplification. These transformations are not endorsed by Madrid or INE.

Explorer and release checks

The observed research explorer has three datasets: 2015–2025 common areas, 2015–2025 whole city and 2014–2024 whole city. Preset indicators are partitioned by spatial level and age selection; profiles are lazy-loaded. The full checksum inventory is a separate downloadable audit artifact.

Dataset	Index + indicators gzip	Typical initial data gzip	Passed
2014-2024-city	732 bytes	20,600 bytes	Yes
2015-2025-areas	291,220 bytes	322,867 bytes	Yes
2015-2025-city	731 bytes	20,791 bytes	Yes

Typical initial data includes the manifest, index, indicators, default map geometry and first area profile. Every artifact and CSV/Parquet download has a verified checksum. Limits remain 500 KB initial data, 1 MB per geometry layer and 100 KB per profile.

Local validation: 70 Python tests pass. Both fixture and observed research browser checks pass, including maps, spatial levels, three research windows, lazy indicators, cohort exclusions, custom ages, keyboard selection and mobile overflow. No research-browser JavaScript errors or failed HTTP responses were observed.

Official statistical release remains blocked by the 2021 discrepancy, historical administrative boundary validity, changed-component analytical approval, and final empirical review/signoffs. Research publication does not open those gates.

Reproduction

Use Python 3.12 and requirements.lock, plus PyShp 2.3.1 for the source geometry reader. Acquire the hash-pinned sources, then run the commands below. The checked-in control responses are hash-bound to their exact request parameters; fetching them again is optional and may reveal source revisions.

```
python -m madrid_demography.cli acquire --lock sources.lock.yml --root data/raw
python scripts/build_research.py
python scripts/check_research_budgets.py
python scripts/check_research_browser.py
```

The website is built in data/derived/research-site. GitHub publication uses the gh-pages branch; repository Pages configuration is a separate hosting setting. The implementation branch retains code, audit evidence, results, PDF and the downloadable website archive.

Official input integration — 2 October 2026

Current observed research calculations, annual reconciliation, sensitivity checks and explorer results are documented in [RESEARCH_RESULTS.md](#). Those calculations do not constitute an official release approval. Earlier acquisition audit sections below retain their recorded snapshot counts; the current lock additionally includes the archived INE reuse terms (41 pinned resources).

The current pipeline now shares a pinned manifest with the empirical audit: 29 immutable snapshots covering January 2014–2025, 2015/2025 section geometries, independent annual controls, municipal methodology and mortality sources. Raw files remain outside Git. `python -m madrid_demography.cli acquire --lock sources.lock.yml --root data/raw` restores the newer pipeline's extensionless, checksum-addressed layout. Audit readers resolve files under the supplied raw root; receipt paths from another machine are not used for pinned inputs.

The provisional boundary audit additionally needs `pip install "[audit-gis]"` (PyShp); the publication GIS importer continues to use GDAL.

New corroboration

The historical catalogue's January 2015 TXT and CSV exports contain the same 232,993 rows, including identical blank cells. This rules out a CSV-specific loss; it does not define blank semantics.

The municipal statistical bank's public monthly query returns exactly the same four sex/nationality totals as the CSV profiles in both endpoint years:

January	Known CSV population	Monthly bank total	Difference
2015	3,149,663	3,149,663	0
2025	3,540,364	3,540,364	0

The responses are retained in `docs/audit/monthly-bank-2015.json` and `monthly-bank-2025.json`; their hashes and comparisons are in `integration.json`. Source:

https://servpub.madrid.es/CSEBD_WBINTER/seleccionSerie.html?numSerie=0301000000001. Its public UI posts JSON to `detalleSerie.html` with `accionFormulario=consultarDatosBarrio`, January, the endpoint year, all districts/barrios/ages, total nationality and total sex, and absolute values. These saved controls corroborate the monthly extraction. They are not revised annual or INE legal population controls.

Known totals matching exhaustive monthly controls is evidence against omitted positive population at city level. It is not an explicit source definition of each of the 304,657 baseline blanks. Audit exports continue to retain nulls; the model-facing normaliser continues to reject blanks.

Pipeline changes

The strict model normaliser can now use a versioned `barrio_code_format`: `district_times_100` (101 becomes 0101; 1002 remains 1002), and explicitly listed `open_age_labels`, including `100 o +`. Existing fixture conventions remain available. Parent conflicts, missing counts, duplicate cells and invalid ages still stop model preparation. The audit normaliser preserves exceptional and split-parent geography for review.

Audit configuration remains separate from `project.example.json`. An executable official model configuration cannot be completed until nullable counts, geography and historical parent contracts have been resolved. No official explorer bundle is published by these changes.

Subsequent blank-count policy

District-by-single-age controls now match all 7,560 four-way count totals for baseline ages 0–89. A conditional research candidate infers 261,837 zeros, preserves raw null status, and leaves 42,820 out-of-scope blanks null. See [BLANK_COUNT_POLICY.md](#) for assumptions, evidence and reproduction. This is not a source definition or official release approval.

Validation and next step

65 automated tests pass. `scripts/check_official_inputs.py` verifies all 29 source hashes and sizes, full CSV/TXT equality, and all four monthly control totals. Endpoint audit exports conserve known counts and null counts; national mortality covers all 2,000 required sex/year/age cells.

Next: review changed geometry components and historical barrio/district parents, alongside the documented monthly-source and conditional blank-cell policy. Annual revised differences remain 7,672 people in 2015 and 12,440 in 2025; the monthly controls show these differences are between published series, rather than a failure to sum the CSV. The exact revisions remain unexplained. Independent survival reproduction, regional mortality sensitivity, and licensing review also remain required before an official release.

Subsequent boundary review: [BOUNDARY_REVIEW.md](#) confirms the geometric components and documents 50 parent exceptions, including 34 unchanged polygons with changed or ambiguous raw parent assignments. Only 94.47% of known 2025 population lies in zones with one unchanged raw barrio code; independent historical parents remain unverified.

Subsequent acquisition: [HISTORICAL_PARENT_SOURCES.md](#) documents independent candidate geometry and the official 1988–2024 section registry. The manifest now has 40 verified snapshots. The registry corroborates 33 of 34 unchanged-geometry parent exceptions; all six geometry candidates remain blocked on endpoint validity and containment.

Endpoint reconciliation and normalisation

Completed 2 October 2026. Independent comparisons and audit normalisation are implemented. **Official analysis remains gated**: the source series differ, baseline blanks remain unknown, and the geographic crosswalk requires review.

Independent published counts

The downloader acquired and pinned INE [table 2881](#), Madrid's municipal methodology, and two historical section/nationality workbooks from the municipal [bank of data](#). The bank tree supplies series IDs 0302010300061 (2009–2019) and 0302010300062 (2020–2026). The workbook sheet labels and titles explicitly identify the revised padrón at 1 January of each endpoint year.

Reference date	Monthly raw known-count sum	Annual revised municipal total	INE legal total	Monthly minus municipal revised	Monthly minus INE
2015-01-01	3,149,663	3,141,991	3,141,991	7,672	7,672
2025-01-01	3,540,364	3,527,924	3,506,730	12,440	33,634

Madrid's methodology says the municipal statistical exploitation is exhaustive but its counts do not have the legal status of the INE figures and can differ from them. The monthly files and annual revised workbooks also differ. These statements identify distinct series; they do not establish the cause of each individual difference. **No counts are rescaled or overwritten to force agreement.**

The [reconciliation report](#) retains both comparisons by sex, the revised four-category counts, every differing published section total, unmatched codes and source hashes. Published section totals sum exactly to the revised city totals. Differences across sections, raw-only geography and uncoded residents sum exactly to the city discrepancy. The 2015 workbook differs from the monthly extract in 2,222 of its 2,420 sections; the 2025 workbook differs in 2,331 of 2,462. This is a broad revision difference, not an isolated missing section. The precise revision process and the choice of analytical source series still need to be settled.

The municipal workbooks have a separate **No consta** nationality group: 16 people in 2015 and 139 in 2025. Including this group closes the published identity Spanish + foreign + No consta = total, including the sex totals. For comparisons against raw EXT, foreign and No consta are combined because the raw-source documentation explicitly includes unknown nationality in EXT. This category mapping is recorded; the report does not equate raw EXT with the workbook's foreign-only column.

Correction to the preliminary profiler

The initial profiler skipped rows with invalid geography before summing counts. Normalisation exposed two uncoded-section rows in 2025, containing two people in district 06, barrio 0606, at ages 42 and 50. Those rows are now retained in the city counts and flagged as missing geography. The corrected raw 2025 total is **3,540,364**, two above the earlier preliminary number. Five people in uncoded 2024 rows are also retained, correcting that known-count sum to **3,473,648**. Profiles and geographic population coverage were regenerated.

No artificial section code is assigned. The 2025 endpoint has five known people without a matching polygon: two without section codes and three in 2807908130, absent from the geometry archive. The 2015 endpoint has twelve known people in exceptional codes without polygons. Synthetic tests now ensure an invalid geography cannot silently remove population from the audit sum.

Explicit input contracts

[config/schema_aliases.json](#) pins exact column aliases, encodings and delimiters for both endpoint files. Unexpected headers, negative counts, duplicate full-grain source rows, inconsistent codes and mismatched embedded reference dates fail normalisation.

The generated long-form records preserve reference date, vintage section code, district, barrio, age label and kind, sex, nationality, nullable population, cohort eligibility, geometry presence, source hash, schema version and raw row number. The full grain includes barrio membership. The two age-45 rows for section 2807919036 remain separate rather than being dropped or assigned one parent.

The [normalisation report](#) verifies:

- 2015: 931,972 long-form cells; 3,149,663 known people conserved; all 304,657 blank count cells retained as unknown.
- 2025: 962,528 long-form cells; 3,540,364 known people conserved; no blank count cells.
- Invalid ages, open terminal groups, births during the interval, missing geometry and split barrio membership remain visible rather than disappearing during conversion.

An empty population field means **unknown**, not zero. These audit records deliberately do not fit the production model's non-null, harmonised input contract. They cannot be used as official model inputs until that contract is satisfied.

Mortality and terminal ages

[config/analysis.json](#) adopts Spain's national annual single-age mortality from table 27153, with an explicit caveat that national period tables are not Madrid-specific. The normaliser verifies source metadata units and complete coverage, then exports all **2,000 male/female age/year probabilities** at ages 0–99 for 2015–2024. It uses *Riesgo de muerte / 1000*; it never substitutes the death-rate measure or expands the open terminal group.

The paired counterfactual covers **endpoint ages 10–99**, corresponding to baseline ages 0–89. Baseline ages 90+ and endpoint ages 100+ are retained as separate observed/excluded records. This avoids assigning invented single ages to 100 o + or projecting into an unsupported terminal-age mortality schedule. The anomalous baseline age 135 remains an exception. Endpoint ages 0–9 remain observed-only.

Known population partitions reconcile to the full known sums:

Endpoint file	Eligible cohort known count	Observed births during interval	Terminal-age known count excluded	Invalid-age known count excluded
2015 baseline	3,109,701	—	39,961	1
2025 endpoint	3,275,658	262,476	2,230	0

Regional mortality sensitivity remains pending. This narrower age scope is explicit in the configuration and must accompany any eventual results.

Reproduce and remaining release work

After acquiring the pinned sources and generating profiles/boundaries, run:

```
PYTHONPATH=src python -m madrid_demography.reconcile
PYTHONPATH=src python -m madrid_demography.normalise
PYTHONPATH=src python -m madrid_demography.audit
```

The XLSX reader uses the Python standard library and reads stored workbook values, not recalculated formulas. Normalisation verifies pinned raw hashes and conserves both known counts and unknown cells. Generated CSVs stay under ignored data/normalised/; tracked reports contain their hashes. Production remains blocked until a compatible monthly/revised source policy and blank-count rule are supported by evidence, geographic components and historical parent boundaries are reviewed, and release QA passes. The demonstration website remains unchanged.

Subsequent work: [the conditional blank-count policy](#) uses matched district/age controls to produce a labelled research candidate for eligible 2015 cells. Raw audit blanks remain null, and the policy does not approve an official release.

Official source acquisition and preliminary feasibility audit

Current observed research calculations, annual reconciliation, sensitivity checks and explorer results are documented in [RESEARCH_RESULTS.md](#). Those calculations do not constitute an official release approval. Earlier acquisition audit sections below retain their recorded snapshot counts; the current lock additionally includes the archived INE reuse terms (41 pinned resources).

Audit date: **2 October 2026**. Outcome: **sources available; official analytical release remains blocked**. The analytical pipeline and static demo already exist. This audit does not replace the demonstration payload with official results.

This document records the original acquisition snapshot. Subsequent [integration](#), [conditional count-policy](#) and [boundary review](#) reports contain the current findings.

Acquired evidence

All **28 source snapshots** were downloaded, inspected and pinned by SHA-256 in [sources.lock.yml](#). The manifest records exact URLs, retrieval dates, explicit CSV encodings and delimiters, and reference-date evidence. Pins approve acquisition snapshots only. Raw files and unique receipts are retained locally under ignored data/raw/; the tracked [acquisition report](#) now records verification against the pinned manifest; the local receipts retain final URLs, timestamps, response metadata and resource discovery. Raw bytes are excluded from Git.

The Madrid portal has migrated. The plan's legacy 200076-0-padron.xml and .csv endpoints returned HTTP 404. The live catalogue links to a separate [historical dataset](#), whose [RDF resource catalogue](#) exposes January resources for every year from 2014 through 2025. Resource filenames have numeric identifiers that do not encode the year; the manifest uses catalogue descriptions rather than guessed filenames.

The downloaded [official content/structure PDF](#), version October 2019, explicitly states: “La fecha de referencia de cada fichero mensual de un mes genérico ‘m’ es el día 1 de dicho mes ‘m’.” January files therefore refer to 1 January. Recent files also contain reference-date and load-date columns; these are preserved separately in profiles. The PDF notes that EXT includes stateless people and people with nationality not recorded.

Population profiles

The [machine-readable profiles](#) retain headers, age distributions, row counts, known count sums, blank-cell counts, geography inventories, conflicts and recent date values. Encoding varies between UTF-8 and Windows-1252. Field casing, quoting and whitespace also change between vintages.

Reference date	Rows	Raw section codes	Sum of nonblank population counts	Blank count cells
2014-01-01	232,074	2,417	3,176,508	296,007
2015-01-01	232,993	2,424	3,149,663	304,657
2016-01-01	233,469	2,420	3,174,945	302,903
2017-01-01	233,949	2,421	3,191,117	300,262
2018-01-01	236,337	2,443	3,231,062	297,248
2019-01-01	236,891	2,443	3,275,195	289,584
2020-01-01	237,581	2,443	3,345,894	280,036
2021-01-01	237,758	2,445	3,292,949	0

Reference date	Rows	Raw section codes	Sum of nonblank population counts	Blank count cells
2022-01-01	238,221	2,445	3,296,033	279,395
2023-01-01	238,497	2,451	3,352,448	0
2024-01-01	239,074	2,451	3,473,648	0
2025-01-01	240,632	2,463	3,540,364	0

These are **raw known-count sums**, not independently reconciled official city totals. Blank cells are counted as unknown and excluded from the known sum, without treating them as observed zero. No inference about suppression or zero encoding has been approved.

Actionable schema findings:

- The 2015 file has one recorded age of 135, outside the existing analytical contract; preserve it in an exception table and obtain an explicit treatment.
- Starting in 2023, 100+ is an open age group. The 2025 endpoint contains 1,346 rows with this label. The pipeline currently expects integer ages; it needs a declared terminal-cohort policy before these sources can be normalised.
- Four exceptional 2015 section codes have no polygon: 2807901888, 2807912999, 2807915999, 2807919999, containing 12 known people in total. They must remain visible in reconciliation and exception accounting.
- The 2025 source includes 2807908130, with three people, absent from the 2025 geometry archive, plus two people without section codes. All five remain in city accounting. The earlier profiler skipped uncoded rows; this is corrected in the follow-up report.
- Section 2807919036 has two barrio memberships in the 2025 source. At age 45 it has two separate rows: 14 people in barrio 1902 and one in barrio 1901. This is a duplicate at section-age grain, not a byte-identical duplicate. Do not drop either row or choose a parent silently.
- Intermediate-year totals show changes that require administrative/revision checks, especially 2020–2021 and 2023–2024. A change in the raw sum alone does not identify its cause.

Mortality coverage

The plan's TABLA/27154?tip=AM URL returns HTTP 200 with the text “La operación indicada no existe (TABLA)”, rather than JSON. The downloader rejects it. Working sources are the official CSV export and SERIES_TABLA metadata endpoint, now recorded in the manifest.

The [mortality report](#) establishes:

- **Madrid regional table 27154:** annual observations cover the required 2015–2024 period, but ages are grouped: 0, 1–4, five-year groups and terminal groups. It does not supply the 2,000 sex/year/single-age cells needed at ages 0–99. Missing observations in redundant terminal categories are recorded separately. Grouped risks must not be passed to the existing single-age model.
- **Spain national table 27153:** all 2,000 male/female single-age risk cells at ages 0–99 across 2015–2024 are present, with an additional 100 y más años terminal group. This is a viable source candidate if national mortality is explicitly adopted; it does not establish small-area mortality accuracy.
- Both metadata sets label Riesgo de muerte in **Tanto por mil**. If adopted, divide that measure by 1,000 to obtain qx. Tasa de mortalidad is a distinct measure and must not be substituted. CSV numeric parsing must account for decimal commas and period thousands separators.

The follow-up normalisation configures national single-age mortality with an explicit caveat and paired endpoint ages 10–99. Regional sensitivity remains pending. See [endpoint reconciliation](#) for the implemented policy; grouped regional risks are not interpolated.

Preliminary geographic feasibility

Both official archives downloaded successfully. Their layer names identify 1 January 2015 and 1 January 2025; both declare ETRS89 / UTM zone 30N, verified as EPSG:25830. The Madrid municipality contains **2,420** start-year and **2,462** endpoint polygons. No Madrid polygon is empty or invalid. The legacy GIS landing page in the original plan describes November 2011 cartography; it is not reference-date evidence for these archives.

The [spatial report](#) records every bipartite overlap component and its old/new codes, mutual overlap and symmetric-difference area. At the plan's provisional 99.9% mutual-overlap tolerance:

- 2,221 components are candidate unchanged sections with matching IDs.
- 122 components are candidate stable aggregates of changed sections. The largest contains 19 sections in one vintage.
- Candidate comparable components cover **99.9996%** of known 2015 counts and **99.9999%** of known 2025 counts. This exceeds the proposed 95% feasibility threshold provisionally; the denominator includes known counts only and does not resolve older blank cells.
- The whole-municipality symmetric difference is approximately 7.59 m². Aggregate internal overlap is negligible in 2015 and approximately 0.082 m² in 2025, subject to floating-point precision.

The graph excludes intersections at or below the larger of 1 m² and 0.1% of the smaller polygon area; excluded intersections total approximately 41.23 m². These settings and all candidate labels are explicit in the report. They are an audit heuristic, not approved population-allocation weights. No polygons were repaired, clipped or interpolated.

The spatial gate remains open: changed components and borderline differences need review; every ignored materiality choice must be acceptable; missing source codes need resolution; historical barrio/district geometries and parent containment still require independent verification. A high component coverage percentage does not approve current barrio boundaries or resolve the source's multiple-barrio section.

Remaining production gates and next implementation

The [feasibility report](#) keeps `production_ready=false`. Independent INE and revised municipal totals are now acquired, differences are accounted for by section, and audit normalisation is implemented; see [the follow-up report](#). The original production checklist remains:

1. Obtain independently published endpoint totals and reconcile differences, including blanks and exceptional codes. Then check barrio/district totals and intermediate-year administrative breaks.
2. Define versioned raw-schema aliases and explicitly approved blank, unknown-age and terminal-age rules. Retain exception rows and audit totals; resolve section/barrio membership without loss of counts.
3. Adopt and configure a documented mortality method and terminal-cohort treatment; validate selected survival chains and sensitivity to region.
4. Review changed spatial components and historical parent boundaries, then produce a versioned crosswalk with conservation checks and approved reliability labels.
5. Run the normalised official pipeline and full release QA. Replace the demo only after those gates pass.

Reproduce

Use the commands in [README.md](#). The acquisition step fails on checksum drift, invalid payloads, truncation, oversized downloads or HTTP/network errors. Each attempt retains a unique receipt. Metadata snapshots can change independently of data; inspect changes before updating a pin. The report builder verifies stored bytes against both receipts and manifest pins. The population profiler preserves uncertainty rather than repairing source semantics.

The spatial audit's optional dependencies are pinned in `pyproject.toml`; analysis and acquisition remain standard-library Python. The executed GIS environment used Python 3.12, Shapely 2.1.2, pyshp 2.3.1, pyproj 3.7.2 and NumPy 2.3.5. Synthetic checks cover exact splits, renumbering and same-ID redraws; acquisition checks cover checksum drift, immutable repeated retrieval, size limits, truncation, malformed responses and

network failure.

Boundary and historical parent review

The 2015–2025 section overlay supports stable geographic aggregates, but it does **not** establish comparable barrio totals. Raw historical membership reveals 50 parent exceptions. Official small-area publication remains blocked on independent historical parent boundaries and analytical review.

Geometry findings

The review uses both pinned INE archives in EPSG:25830 and the actual publication pipeline's overlay. District and section attributes (CDIS, CUDIS, CSEC, CUSEC) agree throughout both archives. Polygons are valid, and both vintages pass the pipeline's within-vintage overlap audit.

Component class	Components
Unchanged section geometry	2,221
Splits	60
Merges	47
Many-to-many redraws	15
Total	2,343

All 122 changed component unions agree within tolerance. The maximum symmetric-difference ratio is **1.43×10^{-7}** , well below the 0.001 criterion. The component sets are identical to the earlier provisional audit even with the pipeline's smaller material-overlap threshold (10^{-6} of the smaller polygon's area). The code calls many-to-many components `materially_redrawn`; in these 15 cases that describes internal section rearrangement, while the common outer union remains compatible with exact aggregation.

Population accounting conserves all known city counts: 12 people in exceptional 2015 section codes lack polygons; in 2025 three people lack a polygon and two lack section codes. These exceptions remain separate from mapped section results.

Parent findings

Parent assignments are read from each endpoint's full raw barrio/section grain, including zero and blank count rows. A majority-count barrio is never selected to resolve ambiguity. Matching raw codes provide corroboration only; they do not replace an independently dated barrio boundary layer.

Parent status	Components
One matching raw parent code at both endpoints	2,293
One raw barrio reassigned between endpoints	41
Component spans multiple barrios	8
Section has multiple raw barrio memberships	1

Of the 50 exceptions, **34 have unchanged section geometry**. Examples include section 2807918047, whose raw barrio changes from 1801 to 1803, and sections in district 19 changing among 1901, 1902 and 1903. The polygons alone cannot resolve these changes.

Section 2807919036 has one person assigned to barrio 1901 and 2,058 assigned to 1902 in 2025. Both raw memberships remain represented.

One common component, Z-57b172e30c85, spans districts 16 and 21 and five barrio codes. It combines eight old and nineteen new sections, with 14,912 known people in 2015 and 40,929 in 2025. Assigning this entire stable

zone to a single current district or barrio would change the hierarchy.

Endpoint	Known population in zones with one matching raw barrio code	Share of known city population
2015	3,029,099	96.17%
2025	3,344,556	94.47%

The earlier >99.99% candidate coverage concerned section geometry alone. The 94.47% figure is a narrower raw-parent feasibility screen, not an approved coverage measure and not a replacement definition of the plan's geometry gate. It shows why geometry coverage cannot justify publishing comparable barrio totals.

Review artifacts and reproduction

The [seven-page atlas](#) shows all 122 changed components plus the 34 unchanged components with parent exceptions. Blue outlines are 2015 and orange outlines are 2025. Codex inspected all seven contact-sheet pages; the outer-union coincidence and internal split/merge patterns are consistent with the numerical results. This visual inspection is not an independent verification of historical parent boundaries or an analytical release approval.

The full inventory is in [audit/boundary-review.csv](#) and [audit/boundary-review.json](#), including source hashes, land area, known population, parent codes, geometry metrics and accounting exceptions. Atlas hashes and plotted component IDs are in [audit/boundary-atlas.json](#). Audit tables deliberately use separate column names from the publication crosswalk.

```
pip install '[audit-gis,audit-plots]'  
python scripts/review_official_boundaries.py  
python scripts/render_boundary_atlas.py
```

58 tests pass. Parent tests cover reassignment despite identical section IDs, cross-district zones, ambiguous membership, unknown codes, missing evidence and conservation of exceptional population rows. Ruff lint and formatting pass.

Next step

Acquire independently dated Madrid barrio and distrito boundaries, or official change records sufficient to construct and validate common historical parents. Start with districts 18 and 19 and the cross-district component above. Overlay those parents against both section vintages and retain explicit exceptions. Current boundaries or raw-code dissolves cannot serve as independent 2015 boundary evidence.

City-level research calculations can proceed without assigning a barrio to these components, provided they include exceptional population and retain the conditional count and mortality caveats. They would still need independently reproduced survival and the remaining release reviews before official publication.

Follow-up: [independent parent sources and change records](#) are now acquired. The registry corroborates 33 of 34 unchanged-polygon parent exceptions. Candidate polygon layers fail endpoint-validity/containment requirements; no independent parent geometry has been approved.

Independent historical parent sources

Independent municipal parent geometry and official section-change records are now acquired and pinned. The registry corroborates 33 of the 34 unchanged-geometry parent exceptions. **No endpoint parent geometry is approved:** candidate layers have temporal uncertainty and substantial containment discrepancies against the INE section cartography.

Acquired sources

The official CKAN catalogue exposes [municipal barrios](#), [administrative limits](#), [dated birth maps](#), and [section-history documentation](#). Exact download URLs, hashes and sizes are pinned in the source lock. The manifest now contains **40 verified snapshots**.

The downloaded historical-division ZIP contains eleven nested archives, from 1612 through **1987**. Its newest division has 128 barrio polygons and 21 district polygons in EPSG:25830. The 1987 label does not certify January 2015 validity.

The live barrio/district downloads contain 131 and 21 polygons. The pinned Geoportal metadata explicitly documents a municipal boundary alteration dated **22 October 2025** and a further adjustment in **2026**. Those live layers cannot be substituted for January 2025.

The 20250101_nacimientos.zip statistical map archive contains 131 barrio and 21 district polygons. Its shapefile members are dated **2 October 2025**, before the October municipal alteration. That strengthens its candidacy compared with the live service, but a statistical filename and ZIP timestamp are not an explicit January boundary-validity interval. Birth counts in this archive are not used in the demographic model.

Independent containment checks

All six candidate layers were read with explicit field/encoding contracts, normalized parent codes and EPSG:25830 validation. The reader rejects invalid/empty/duplicate polygons, unexpected feature counts and encoding/CRS drift. No geometry repairs were performed.

Each INE section was compared with the union of its raw-declared parents. Multiple raw memberships remain explicit; the largest group is not selected. The containment threshold allows at most 0.1% of a section's land area outside its declared parent.

Candidate layer	Endpoint tested	Sections outside declared parent at strict tolerance
Historical 1987 barrios	2015	926 / 2,420
Historical 1987 districts	2015	354 / 2,420
January-2025-labelled statistical barrios	2025	932 / 2,462
January-2025-labelled statistical districts	2025	340 / 2,462
Live 2026 barrios	2025	939 / 2,462
Live 2026 districts	2025	349 / 2,462

Many failures may concern different precision or adjustments to road axes, but not all are tiny. In the January-2025-labelled layers, **50 barrio cases and 20 district cases exceed 5% outside area**. Section 2807910217 has approximately 42.4% outside its declared barrio and district. The older historical layer puts approximately 98.7% of the 2015 polygon outside its declared parent. These are measured map incompatibilities; they do not show that the people or raw parent codes are wrong. Increasing tolerance or assigning the nearest/majority parent would conceal the problem.

Full source contracts, labels, layer timestamps, population accounting and exception metrics are in [parent-candidates.json](#) and [config/parent-candidates.json](#). Temporal validity remains false for all six candidates. The parent-vintage guard rejects missing, malformed or out-of-range date evidence.

Official section-change registry

The municipality's historical workbook contains **2,713 records**, titled “Historia del seccionado 1988 - 2024”, with creation, deletion and modification fields, previous barrio codes, origin/destination sections and notes.

Using creation-inclusive/deletion-exclusive intervals for this audit:

- All **2,420 ordinary 2015 section memberships** match. The four exceptional no-polygon codes remain outside the registry.
- **2,460 raw 2025 section memberships** match. Section 2807908130 is absent from the registry; 2807919036 has raw memberships in both 1901 and 1902, while the registry records 1902; and 2807919052 lacks a creation date in registry row 2546.
- The workbook documents **54 parent reassignments** and corroborates **33 of the 34** unchanged-polygon parent exceptions in the boundary review. The remaining unchanged-polygon case is the split membership of 2807919036.

23 modification-date fields contain 20171606, invalid as YYYYMMDD. They are preserved verbatim. Where a previous barrio is recorded, only the year is used to bracket endpoints strictly before or after 2017; no June/October date is invented. Within-year comparisons would remain unresolved. Carrying a workbook titled through 2024 into January 2025 is provisional, even where the raw memberships agree.

These records help explain historical membership changes, but do not reconstruct exact polygon boundaries or locate individual people within split sections. Evidence, preserved date errors and all discrepancies are in [section-history.json](#).

Reproduce and continue

```
pip install '[audit-gis,audit-history]'
python -m madrid_demography.cli acquire --lock sources.lock.yml --root data/raw
python scripts/audit_parent_candidates.py
python scripts/audit_section_history.py
```

65 tests pass, including temporal validity, malformed dates, partial containment, missing and ambiguous parents, and registry dates that must not be silently repaired. Ruff lint and formatting pass. All 40 pinned source hashes and byte sizes were verified.

Next, build a **city-level research calculation** using the conditional count policy and national mortality, including exceptional population. This can test the survival model while the spatial publication gate remains closed. Barrio publication still needs January-valid parent cartography or coordinate-based official change records, reconciliation with the section geometry, and the remaining analytical release reviews.